NovaCare Responsible AI Framework

Integrated final framework for governing AI across its full lifecycle

NovaCare Health Systems

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Assignment: 6.5 Final RAI Framework

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NovaCare Health Systems

Fictional healthcare enterprise used throughout the course

- **Scale** — 30 hospitals, 150+ outpatient facilities, approximately 45,000 employees
- **Reach** — About 6 million patients a year through hospitals, clinics, diagnostic centers, and telehealth
- **Technology** — Enterprise EHR, cloud analytics, imaging, patient portals, and digital care platforms
- **Why RAI matters** — AI errors or bias can affect patient safety, privacy, treatment decisions, and trust

AI Use Cases and Risk Tiers

Risk tier sets the level of review, approval, and monitoring

AI-Assisted Medical Imaging

Detects abnormalities in X-rays, CT scans, and MRIs

Tier 1 (high risk)

Patient Deterioration Prediction

Predicts patients at risk of rapid clinical decline

Tier 1 (high risk)

Generative AI Clinical Assistant

Summarizes records and drafts clinical documentation

Tier 2 (moderate risk)

AI Scheduling & Resource Optimization

Forecasts demand and optimizes staffing and appointments

Tier 3 (lower risk)

Framework at a Glance

How the required elements are organized in this deck

Element	What it contains	Slides
Ethical foundation	Ethics statement; seven principles with commitments and owners	5-6
Governance and roles	Oversight structure, seven roles, approval and vetting rules	7-9
Implementation	12-month timeline with checkpoints; employee RAI training	10-11
Metrics and lifecycle	Core KPIs, added metrics, eight-stage lifecycle with disposition	12-14
Risk guidelines	Explainability, data bias, privacy, protection, and compliance	15-17
People and impact	User education, long-term impacts, best practices	18-20
References	Thirteen prioritized APA sources	21

Ethical Principles Statement

Carried forward from the NovaCare ethics statement (Assignment 1.3)

NovaCare Health Systems will use AI only to improve patient care, safety, and access. We keep clinicians and patients in control of consequential decisions. We test AI for fairness across the patients we serve, protect patient data, and explain how AI contributes to care in plain language. Every AI system has a named owner who is accountable for its outcomes, and we can pause or shut down any system that is unsafe. Buying AI from a vendor never transfers our responsibility.

Each commitment is converted into a measurable target, an owner, and a corrective action on the next slides.

Seven Principles: Commitments and Owners

Each principle has one measurable commitment and one accountable role

Principle	Measurable commitment	Owner
Patient Safety & Reliability	No severe AI-related incidents; at least 95% validation pass rate	AI Safety & Validation Lead
Fairness & Equity	Subgroup performance gap of 5 percentage points or less	RAI Governance & Ethics Lead
Privacy & Security	Zero critical findings; 100% of findings closed on time	Data Governance & Privacy Lead
Transparency & Explainability	100% of high-risk models have a current model card	Human Factors & Clinical AI Lead
Human Oversight	Clinician override rate stays within the 5%-20% review band	Human Factors & Clinical AI Lead
Accountability	100% of AI systems have a named owner and a current review	Chief Responsible AI Officer
Governability	Unsafe critical AI disabled within 30 minutes of a confirmed trigger	AI Security & Red Team Lead

RAI Governance Structure

Oversight runs from the Board to each AI system owner, organized by NIST AI RMF (Govern, Map, Measure, Manage)

Level	Body	Decision rights
Board	Board Risk / Quality Committee	Reviews RAI effectiveness, material incidents, and resources once a year
Executive	Executive RAI Steering Committee	Approves Tier 1 systems and re-approves them every 12 months; reviews portfolio risk quarterly
Lead	Chief Responsible AI Officer	Owns the framework; approves Tier 2 systems; can pause any AI system
Control functions	Governance & Ethics, Risk & Compliance, Safety & Validation, Data Privacy & Governance, AI Security & Red Team	Independent challenge; Risk & Compliance Manager approves Tier 3 systems
Advisory	Clinical / Human Factors Advisory Council	Advises on workflow impact, usability, and patient and clinician feedback
System	Business or clinical owner for each AI system	Accountable for outcomes, KPIs, and corrective actions

Key Roles and Responsibilities

Seven RAI roles, each tied to the KPI it answers for

Role	Core responsibility	KPI answered
Chief Responsible AI Officer	Enterprise accountability, policy, executive and Board reporting	Named owner coverage
RAI Governance & Ethics Lead	Ethics reviews, policy interpretation, stakeholder analysis	Subgroup performance gap
AI Risk & Compliance Manager	Risk assessments, HIPAA and EU AI Act tracking, vendor risk, audit readiness	Privacy and security findings closed
AI Safety & Validation Lead	Independent validation, robustness testing, drift analysis	Critical incidents; validation pass rate
Data Governance & Privacy Lead	Data lineage, consent, minimization, retention, de-identification	Privacy and security findings
AI Security & Red Team Lead	Threat modeling, adversarial testing, incident response, kill-switch tests	Time to disable unsafe AI
Human Factors & Clinical AI Lead	Workflow fit, explainability, training, clinician and patient feedback	Model cards; override rate

Oversight Processes

Internal development and vendor purchases follow different paths under one standard

Process	Steps	Rule
Internal vetting	Intake and owner, risk tier and impact map, data and privacy assessment, build and documentation, independent validation, RAI approval, deploy and monitor	High-risk AI cannot self-approve; RAI review decision within 15 business days (proposed)
Vendor vetting	Business need and risk class, due diligence, contract safeguards, independent NovaCare testing, approval and pilot, ongoing monitoring	Procurement does not transfer accountability to the vendor
Review cadence	Weekly model monitoring, monthly owner and RAI review, quarterly Steering Committee, annual Board review	Every KPI has an owner, a threshold, and a cadence
Escalation	Measure, compare to threshold, assign action, verify closure	Critical safety, privacy, or security triggers automatically pause the system

12-Month Implementation Timeline

Phased rollout with an accountability checkpoint at months 3, 6, 9, and 12

Phase	Key milestones	Accountability checkpoint
Months 1-2: Foundation	Executive sponsor named; RAI charter signed; 100% of AI systems inventoried and risk-tiered by day 45	Month 3: charter and inventory reviewed by the Steering Committee
Months 3-4: Organization	All seven roles filled by month 4; Steering Committee formed; RACI and escalation paths set	Month 3: role appointments confirmed to the Board
Months 5-6: Controls	Internal and vendor vetting standards, impact assessments, model card templates, approval gates	Month 6: controls approved by the CRAIO
Months 7-8: Pilot	Framework applied to 2-3 use cases (imaging and deterioration first); training delivered; dashboards and incident process tested	Month 8: pilot review of KPIs and incidents
Months 9-10: Scale	Governance extended to all AI projects and vendors; procurement, security, privacy, and audit integrated	Month 9: Steering Committee gap review
Months 11-12: Optimize	Audit of framework effectiveness; KPI trend review; annual RAI report and Year-2 roadmap	Month 12: Board Risk / Quality Committee review

Employee RAI Training

Training scales with exposure to AI; completion is tracked as a KPI

Audience	Training content	Standard (proposed targets)
All employees	RAI principles, approved AI tools, no PHI in public AI tools, how to report concerns	Onboarding within 30 days of hire; refresher every 12 months
Clinicians and AI users	How each system works, its limits, when to override, automation bias, documentation duties	Completed and acknowledged before first use; refresher within 30 days of a material system change
Builders and product owners	Bias testing, model cards, data lineage, privacy by design, approval gates	Completed before submitting a system for RAI review
RAI role holders and leaders	Validation methods, NIST AI RMF, HIPAA and EU AI Act, incident response	At least one certified person per RAI role in year 1
Accountability	RAI duties written into job descriptions and annual reviews	100% training completion tracked quarterly

Core RAI Metrics

Each KPI has a named owner, a threshold, and a predefined corrective action

Principle	KPI	Threshold	Corrective action
Safety & Reliability	Critical AI incidents; validation pass rate	0 severe; 95% or higher pass	Suspend or roll back; root-cause review
Fairness & Equity	Maximum subgroup performance gap	5 percentage points or less	Bias investigation; retrain or restrict
Privacy & Security	AI privacy and security findings	0 critical; 100% closed on time	Block release or remediate
Transparency	High-risk models with current model cards	100%	Escalate missing or expired documentation
Human Oversight	Clinician override and inappropriate automation rate	Override within 5%-20% band	Workflow review and retraining
Accountability	AI systems with named owner and current review	100%	No owner means no production use
Governability	Time to disable unsafe AI after confirmed trigger	30 minutes or less (critical)	Test kill switch and escalation quarterly

Additional Metrics

Added to cover performance drift, explanation quality, patient communication, and training (proposed targets)

Area	Metric	Target and trigger
Performance drift	Accuracy against validated baseline	Within 5 points; retrain after 2 consecutive months of drift
Explainability	Clinicians who can state why a case was flagged	90% or more within 30 seconds in testing
Explanation quality	Explanation accuracy on audited samples	85% or higher
Patient communication	AI-assisted patient messages labeled as AI; complaints	100% labeled; complaints 2% or less per month, else pause feature within 5 business days
Overreliance	Audit of AI-assisted work for automation bias	5% sample; review trigger if override rate exceeds 20%
Training	Required RAI training completed	100%

AI Development Lifecycle and RAI Actions

Eight stages from design to disposition; no stage is skipped for Tier 1 or Tier 2 systems

Stage	RAI actions	Gate or owner
1. Design	Define intended use, affected patients, risk tier, and human oversight design	Use-case owner; impact map
2. Data	Document lineage and consent; minimize and de-identify; test representativeness	Data Governance & Privacy Lead
3. Development	Build model card, limitations, and acceptance criteria; threat model	Product owner; Security Lead
4. Validation	Independent testing for accuracy, subgroup fairness, robustness, explainability, security	AI Safety & Validation Lead
5. Deployment	RAI approval by tier; controlled rollout; clinician training and rollback plan	Steering Committee or CRAIO
6. Monitoring	Track drift, incidents, subgroup results, overrides, complaints; flag accuracy drops over 5 points	Weekly and monthly reviews
7. Sustainment	Retrain after 2 months of drift and at least annually; re-validate and re-approve	Safety Lead; Steering Committee
8. Disposition	Retire when criteria set at approval are met; replacement clears the same gates before the old model is switched off; archive documentation and data per policy	CRAIO sign-off

Risk Guidelines: Transparency and Explainability

Explanation depth scales with the risk to the patient

Risk level	Required explanation	Guideline
High (Tier 1: imaging, deterioration)	Global: model card with purpose, data, limits, subgroup results. Local: case-level reason a patient was flagged	Clinician must be able to state why within 30 seconds; no black-box approval without CRAIO exception
Moderate (Tier 2: clinical assistant)	Model card; sources shown for generated summaries; AI output clearly labeled	Clinician reviews and signs every AI-drafted note; patient messages labeled as AI-assisted
Lower (Tier 3: scheduling)	Plain-language description of what the system optimizes	Staff can see and challenge schedule changes
Monitoring	Audit explanation accuracy and override patterns	Explanation accuracy 85% or higher; review when override rate exceeds 20%

Risk Guidelines: Data Bias

Checked at the data stage and again at validation; three bias sources follow NIST SP 1270

Use case	Main data bias risk	Required mitigation
AI-assisted imaging	Training images skew toward certain populations, scanners, or sites	Test on local, representative data; report accuracy by subgroup
Patient deterioration	Historical records reflect unequal access to care; proxies such as zip code stand in for race or income	Proxy audit; subgroup calibration review before approval
Clinical assistant	Language model reflects biased or outdated clinical text	Red-team prompts across demographic groups; clinician review of outputs
Scheduling	Patterns of missed appointments disadvantage certain groups	Check access and wait times by subgroup
All systems	Systemic, statistical, and human bias (NIST SP 1270)	Fairness gate: 5-point maximum subgroup gap; failing systems are retrained or restricted

Risk Guidelines: Privacy, Data Protection, and Compliance

HIPAA applies to all PHI; the EU AI Act applies where NovaCare serves or sells into the EU

Area	Requirement	Control
Data minimization	Use the minimum PHI needed; de-identify with HIPAA Safe Harbor (18 identifiers)	Privacy impact assessment before data use
Protection	Encrypt PHI at rest with AES-256; restrict access by role	Quarterly penetration tests; PHI redacted from AI logs within 24 hours
Vendors	Business associate agreements; SOC 2 Type II report	Audit rights and incident notice in every contract
Retention and breach	Compliance documents kept 6 years; medical records per state law; breach notice within 60 days	Incident response plan tested annually
EU AI Act	High-risk systems need risk management, data governance, human oversight, and logging; fines up to EUR 15M or 3% of global turnover	Risk & Compliance Manager tracks regulation and audit readiness

User Monitoring, Training, and Education

Clinicians and patients are supported before use and monitored during use

User group	Training and education	Monitoring
Clinicians	Onboarding on each system's limits and override process; signed acknowledgment before first use	Override rates, 5% automation-bias audit, misuse reports
All staff	No PHI in public AI tools; how to report concerns	Access logs; incident reports
Patients	Notice when AI is used in their care or messages; portal explainer in the top 3 languages (proposed)	Complaints at 2% or less of AI-drafted messages per month
Feedback loop	Clinician and patient feedback channel reviewed by the Advisory Council	Monthly review; findings feed retraining and policy

Long-Term Impacts on Employees and Customers

Reviewed annually and reported to the Board

Group	Impact to evaluate	Commitment and measure
Employees	Job change, deskilling, and overreliance on AI	No AI-driven layoffs without RAI review; redeployment and upskilling offered; engagement tracked against baseline
Patients (customers)	Trust, access, and unequal outcomes	Satisfaction gap across groups of 3 points or less (proposed); complaints reviewed monthly
Patient relationships	Loss of human contact in sensitive communication	Human-required communication types defined; empowerment and authenticity checks on AI messages
Society	Public trust and dependency on AI in care	Annual transparency report; continuity and transition plan when a system is retired

Other Best Practices

Additional practices drawn from course readings and cases

Practice	How NovaCare applies it
Treat benchmarks as evidence, not proof	Test on local NovaCare data before trusting vendor or published performance claims
Keep humans in sensitive communication	Define message types that always need a human author or reviewer
Guard against overreliance	Audit 5% of AI-assisted work; train clinicians to challenge AI output
Keep an auditable data provenance record	Ledger of data sources, consent, and changes for every high-risk model
Choose fairness over a vanity metric	Block release when overall accuracy rises but a subgroup gap exceeds 5 points
Plan for dependency and exit	Continuity commitment and exit terms before any system is switched off or replaced

References

Prioritized: governance and regulation first, then bias and ethics, then practice and risk

Priority 1: Governance and regulation

National Institute of Standards and Technology. (2023). Artificial intelligence risk management framework (AI RMF 1.0) (NIST AI 100-1). U.S. Department of Commerce. https://doi.org/10.6028/NIST.AI.100-1

European Parliament & Council of the European Union. (2024). Regulation (EU) 2024/1689 laying down harmonised rules on artificial intelligence (Artificial Intelligence Act). https://eur-lex.europa.eu/eli/reg/2024/1689/oj

Blackman, R. (2022). Ethical machines: Your concise guide to totally unbiased, transparent, and respectful AI. Harvard Business Review Press.

Priority 2: Bias, ethics, and evidence

Schwartz, R., Vassilev, A., Greene, K., Perine, L., Burt, A., & Hall, P. (2022). Towards a standard for identifying and managing bias in artificial intelligence (NIST Special Publication 1270). https://doi.org/10.6028/NIST.SP.1270

Stanford Institute for Human-Centered AI. (2025). The AI Index 2025 annual report. Stanford University.

Vallor, S. (n.d.). An introduction to data ethics. Markkula Center for Applied Ethics, Santa Clara University.

Zhang, K., Khosravi, B., Vahdati, S., & Erickson, B. J. (2024). FDA review of radiologic AI algorithms: Process and challenges. Radiology, 310(1). https://doi.org/10.1148/radiol.230242

Priority 3: Practice, workforce, and risk cases

Sharkey, A., & Sharkey, N. (2012). Granny and the robots: Ethical issues in robot care for the elderly. Ethics and Information Technology, 14(1), 27-40. https://doi.org/10.1007/s10676-010-9234-6

De Freitas, J., Oguz-Uguralp, Z., Uguralp, A. K., & Puntoni, S. (2026). AI companions reduce loneliness. Journal of Consumer Research, 52(6), 1126-1148. https://doi.org/10.1093/jcr/ucaf040

Dell'Acqua, F., McFowland, E., Mollick, E. R., et al. (2023). Navigating the jagged technological frontier: Field experimental evidence of the effects of AI on knowledge worker productivity and quality (Working Paper No. 24-013). Harvard Business School.

Gartner. (2025, March 25). Gartner says CFOs should reset expectations about AI's impact on workforce productivity and headcount [Press release].

OpenAI. (2025). Model spec.

Furze, L. (2023). Digital plastic.